

NumPy Tutorial Problems

Array Indexing & Slicing (T1) and Basic Operations (T2)

T1 — Array Indexing and Slicing

1. Create a 1D array of integers from 0 to 19. Extract every 3rd element starting from index 1.
2. Given `a = np.arange(1, 26).reshape(5, 5)`, extract the middle 3×3 sub-matrix using slicing.
3. From the same 5×5 array, extract the last two rows and first three columns.
4. Reverse a 1D array of 10 elements without using `np.flip()`.
5. Given a 2D array, extract all elements greater than 15 using boolean indexing.
6. Create a 4×4 array and set every element on the border to 0, leaving the interior unchanged.
7. Given `a = np.array([10, 20, 30, 40, 50])`, use fancy indexing to select elements at positions `[0, 3, 4]` in a single call.
8. Given a 2D array of shape (6, 6), select rows `[0, 2, 4]` and columns `[1, 3, 5]` simultaneously (result should be a 3×3 array, not the full cross-product via naive slicing — think about why `a[[0, 2, 4], [1, 3, 5]]` doesn't give you that).
9. Replace all negative values in an array with 0 in-place using boolean masking.
10. Given a 3D array of shape (2, 3, 4), extract a single 2D "slice" (all rows/cols at a fixed depth index).

T2 — Array Basic Operations

1. Create two arrays of shape (3, 3) and perform element-wise addition, subtraction, multiplication, and division.
2. Compute the dot product and the matrix product (`@`) of two 3×3 arrays and explain why they differ for 2D arrays.
3. Given an array of exam scores, compute mean, median, standard deviation, and variance without loops.
4. Demonstrate broadcasting: add a (3,) array to a (4,3) array and explain the shape rule that makes it valid.
5. Given a (5,5) array, compute the sum along `axis=0` and `axis=1` separately, and explain what each represents.
6. Use `np.where()` to replace all even numbers in an array with -1 and odd numbers with 1.

- 7.** Concatenate two arrays vertically and horizontally using `np.vstack/np.hstack` (or `np.concatenate` with `axis`).
- 8.** Given an unsorted array, sort it and also return the indices that would sort it (`np.argsort`).
- 9.** Reshape a 1D array of 12 elements into (3,4), then flatten it back, and explain the difference between `.reshape()`, `.flatten()`, and `.ravel()`.
- 10.** Compute the cumulative sum and cumulative product of a 1D array.